

05 Random Forest Classification

EP

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Random Forest : Classification

For comparison with current classification literature, below I run the richness and Simpson random forest models using classification over regression. The richness and Simpson diversity indices are split based on median, creating two diversity labels: “high” and “low”. This mirrors the approach taken by recent studies in which two groups of samples are matched, e.g. 500 “flood” and 500 “no flood” samples.

```
library(targets)
library(sf)
library(tidymodels)
library(ranger)
library(sf)
library(dplyr)

set.seed(399)
```

Load Data

Read in the final grid model and transform the Simpson scores. This isn’t strictly necessary for Random Forest but allows for direct comparison with the beta regression model.

```
# Uncomment if using pipeline data:

#setwd("~/CSCT_Pipeline")

#grid_model_all <- tar_read(grid_model_all)

#df <- grid_model_all %>%
  #sf::st_drop_geometry()

#If running with downloaded grid data use:

grid_model_all <- st_read("Data/grid_model.gpkg", quiet = TRUE)

df <- grid_model_all %>%
  sf::st_drop_geometry()
```

```

n <- nrow(df)
df$simpson_adj <- (df$simpson * (n - 1) + 0.5) / n

predictors <- c("n_days", "dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenpace")

```

Model Setup : Richness and Simpson

```

#Classification setup
df <- df %>%
  mutate(
    richness_class = factor(if_else(sp_richness >=
                                     median(sp_richness, na.rm = TRUE),
                                     "high", "low"),
                           levels = c("low", "high")),
    simpson_class = factor(if_else(simpson_adj >=
                                    median(simpson_adj, na.rm = TRUE),
                                    "high", "low"),
                           levels = c("low", "high")))

#RF Specification
rf_spec_class <- rand_forest(
  mtry = tune(),
  min_n = tune(),
  trees = 500) %>%
  set_engine("ranger", importance = "permutation",
             respect.unordered.factors = "order", probability = TRUE) %>%
  set_mode("classification")

folds <- group_vfold_cv(df, group = city)

fold_city_lookup <- folds %>%
  mutate(city = purrr::map_chr(
    splits, ~ as.character(unique(assessment(.x)$city)))) %>%
  select(id, city)

#RF Recipe and Workflow
richness_recipe_class <- recipe(
  as.formula(paste("richness_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

richness_wf_class <- workflow() %>%
  add_recipe(richness_recipe_class) %>%
  add_model(rf_spec_class)

simpson_recipe_class <- recipe(
  as.formula(paste("simpson_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

```

```

simpson_wf_class <- workflow() %>%
  add_recipe(simpson_recipe_class) %>%
  add_model(rf_spec_class)

rf_grid <- grid_regular(
  mtry(range = c(2, 8)),
  min_n(range = c(2, 15)),
  levels = 5) #Total of 25 combinations in the grid.

```

Hyperparameter Tuning

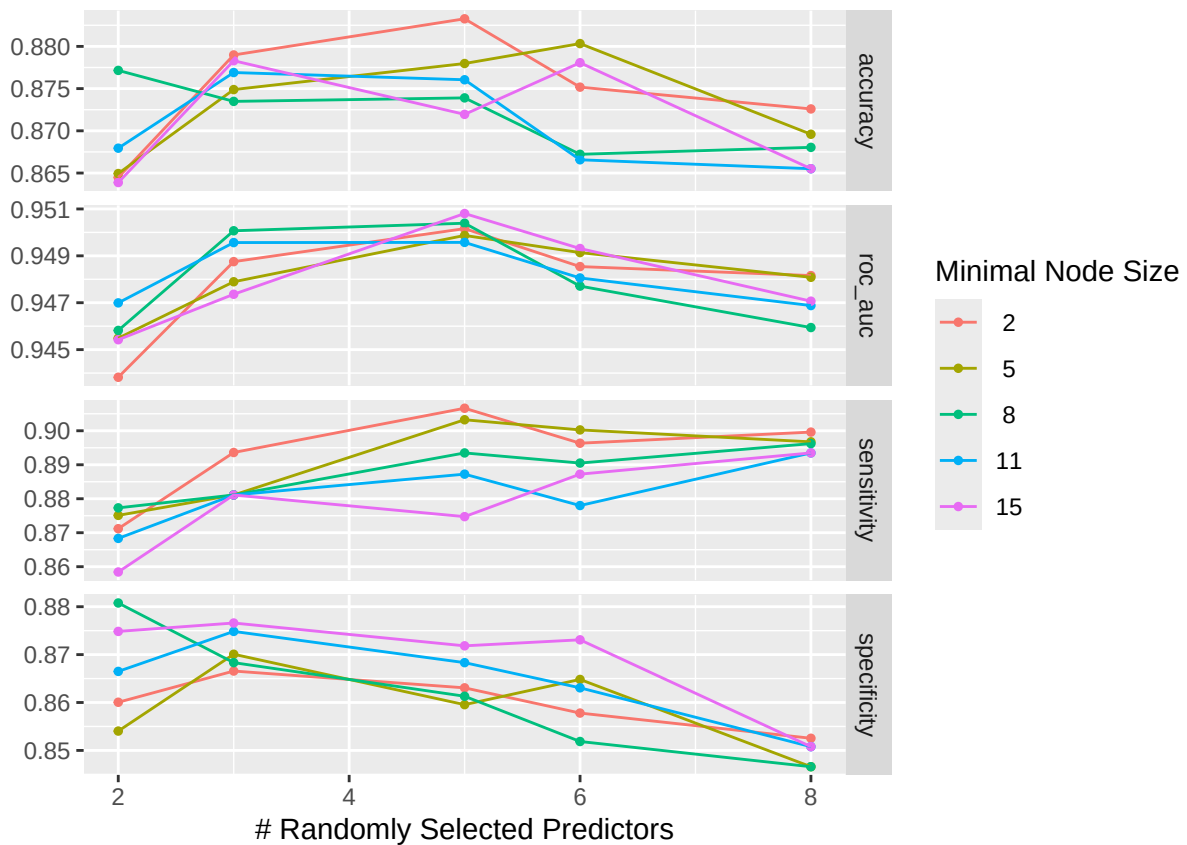
```

richness_res_class <- tune_grid(
  richness_wf_class,
  resamples = folds,
  grid       = rf_grid,
  metrics    = metric_set(roc_auc, accuracy, sensitivity, specificity))

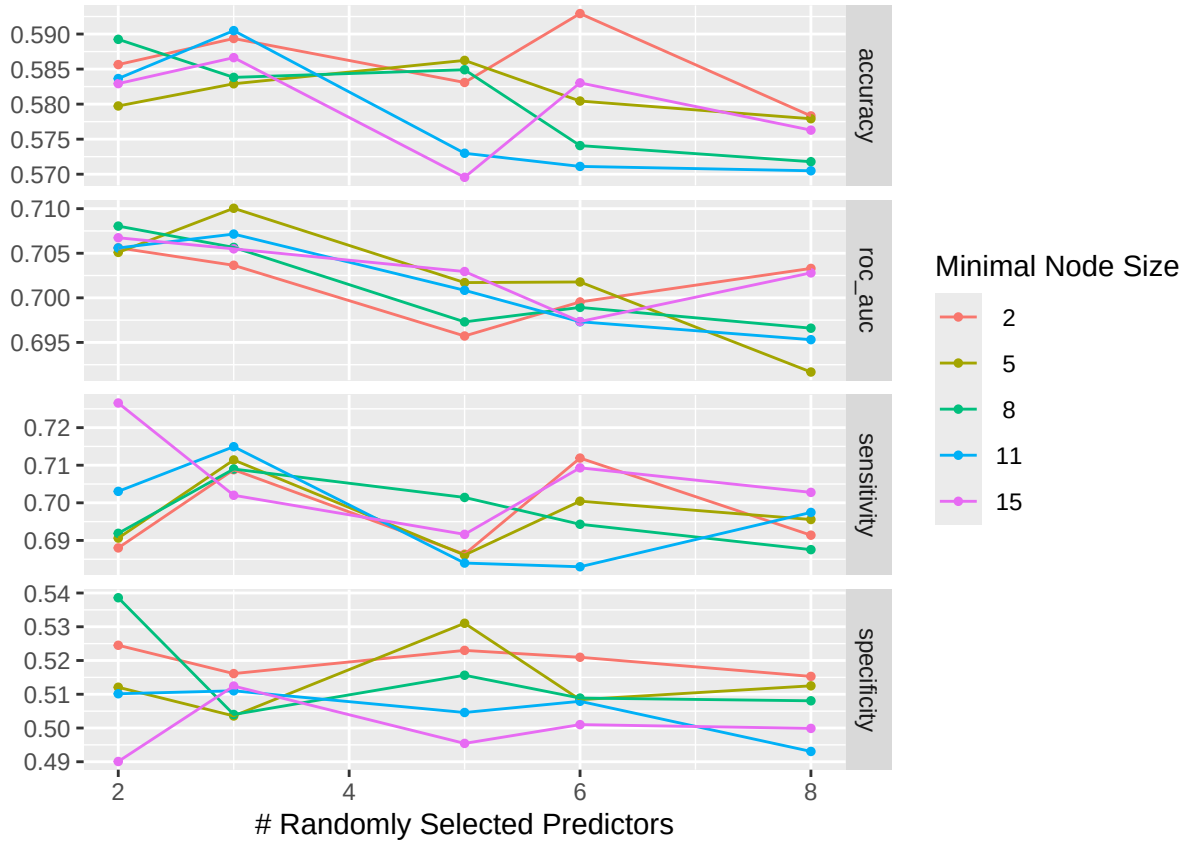
simpson_res_class <- tune_grid(
  simpson_wf_class,
  resamples = folds,
  grid       = rf_grid,
  metrics    = metric_set(roc_auc, accuracy, sensitivity, specificity))

autoplot(richness_res_class)

```



```
autoplot(simpson_res_class)
```



Select Best Hyperparameters

```
show_best(richness_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
  rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
  knitr::kable(caption = "Top 5 hyperparameter combinations:
    species richness class (ranked by ROC AUC)")
```

Table 1: Top 5 hyperparameter combinations: species richness class (ranked by ROC AUC)

mtry	min_n	Mean ROC AUC	Std. error
5	15	0.951	0.010
5	8	0.950	0.011
5	2	0.950	0.011
3	8	0.950	0.013
5	5	0.950	0.012

```
show_best(simpson_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
  rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
  knitr::kable(caption = "Top 5 hyperparameter combinations:
    Simpson diversity class (ranked by ROC AUC)")
```

Table 2: Top 5 hyperparameter combinations: Simpson diversity class (ranked by ROC AUC)

mtry	min_n	Mean ROC AUC	Std. error
3	5	0.710	0.022
2	8	0.708	0.020
3	11	0.707	0.022
2	15	0.707	0.017
3	8	0.706	0.023

```
best_richness_class <- select_best(richness_res_class, metric = "roc_auc")
best_simpson_class <- select_best(simpson_res_class, metric = "roc_auc")
```

The richness model has a best ROC AUC of 0.951, and performance is stable across the grid. This suggests that the model is not highly sensitive to tuning. The Simpson model has an ROC AUC of 0.710 and variability between tuning models. This suggests that the richness model classifies better than the simpson diversity using these predictor variables.

Prediction Rate AUC

This looks at the mean AUC across held-out city folds, and individual fold's AUC scores.

```
# Select best tuning parameters
best_richness_class <- select_best(richness_res_class, metric = "roc_auc")
best_simpson_class <- select_best(simpson_res_class, metric = "roc_auc")

# Overall mean ROC AUC
overall_richness_metrics <- collect_metrics(richness_res_class) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc")

overall_simpson_metrics <- collect_metrics(simpson_res_class) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

overall_richness <- overall_richness_metrics$mean
overall_simpson <- overall_simpson_metrics$mean

# ROC AUC by fold (city)
prediction_rate_richness_byfold <- collect_metrics(
  richness_res_class, summarize = FALSE) %>%
```

```

filter(mtry == best_richness_class$mtry,
       min_n == best_richness_class$min_n,
       .metric == "roc_auc")

prediction_rate_simpson_byfold <- collect_metrics(
  simpson_res_class, summarize = FALSE) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

# Add city names
city_richness <- prediction_rate_richness_byfold %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(richness_auc = .estimate)

city_simpson <- prediction_rate_simpson_byfold %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(simpson_auc = .estimate)

# Combine city-level results
city_auc <- full_join(city_richness, city_simpson, by = "city")

# Add overall ROC AUC
overall_auc <- tibble(city = "Overall",
                      richness_auc = overall_richness,
                      simpson_auc = overall_simpson)

classification_auc_table <- bind_rows(overall_auc, city_auc) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename("City (Held-Out)" = city,
         "Richness ROC AUC" = richness_auc,
         "Simpson ROC AUC" = simpson_auc)

classification_auc_table %>%
  knitr::kable(
    caption = "Overall and city-level cross-validated ROC AUC for species
              richness and Simpson diversity")

```

Table 3: Overall and city-level cross-validated ROC AUC for species richness and Simpson diversity

City (Held-Out)	Richness ROC AUC	Simpson ROC AUC
Overall	0.951	0.710
Nottingham	0.918	0.700
Exeter	0.936	0.775
Newcastle	0.957	0.639
Bristol	0.971	0.705
Gateshead	0.972	0.731

```
# Check for any folds where AUC couldn't be computed (single-class fold)
richness_na_folds <- prediction_rate_richness_byfold %>%
  filter(is.na(.estimate))

simpson_na_folds <- prediction_rate_simpson_byfold %>%
  filter(is.na(.estimate))

richness_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

```
simpson_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

There is lower inter-city variation in the richness model compared to the Simpson model. Lower variation could indicate more consistent performance across held-out cities, however it is hard to compare species richness and Simpson diversity directly since they measure different aspects of biodiversity, on different scales.

Final Model

This is fit on the entire dataset.

```
final_richness_wf_class <- finalize_workflow(
  richness_wf_class, best_richness_class)
richness_fit_class <- fit(final_richness_wf_class, data = df)

final_simpson_wf_class <- finalize_workflow(
  simpson_wf_class, best_simpson_class)
simpson_fit_class <- fit(final_simpson_wf_class, data = df)
```

Success Rate AUC

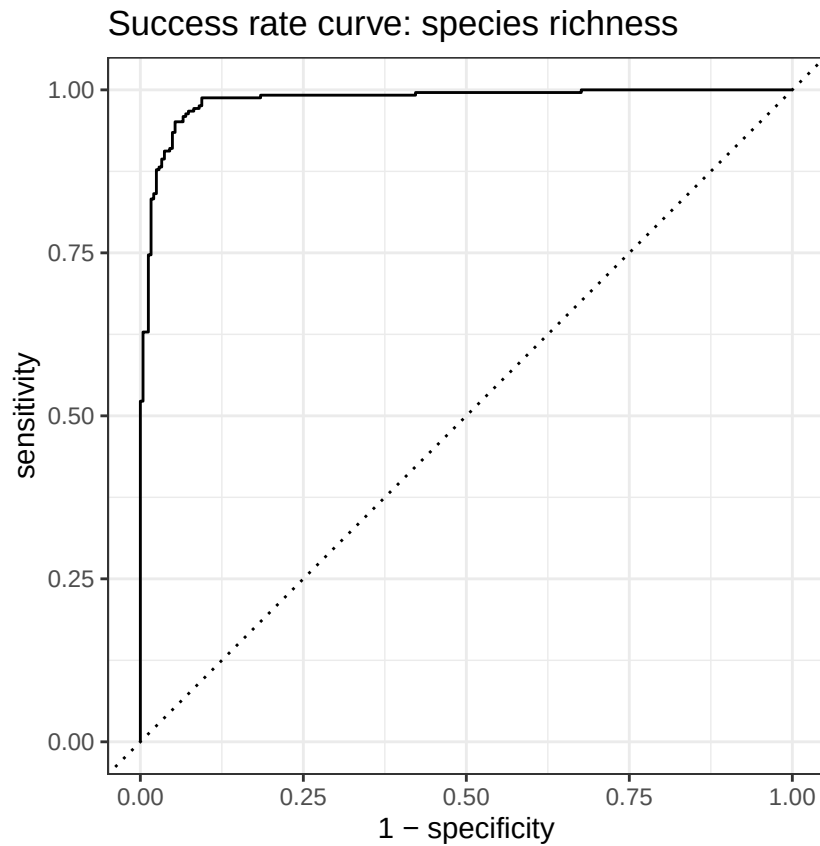
```
richness_success <- augment(richness_fit_class, new_data = df)
simpson_success <- augment(simpson_fit_class, new_data = df)

success_rate_richness <- roc_auc(richness_success, truth = richness_class,
  .pred_high, event_level = "second")

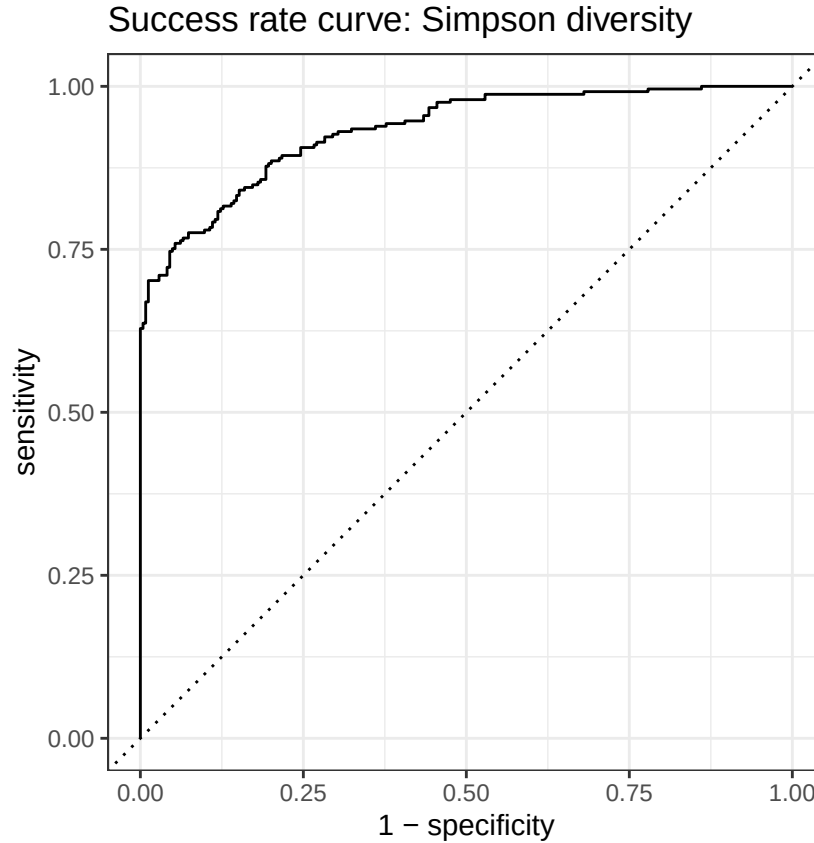
success_rate_simpson <- roc_auc(simpson_success, truth = simpson_class,
  .pred_high, event_level = "second")
```

ROC Curves

```
roc_curve(richness_success,  
          truth = richness_class,  
          .pred_high,  
          event_level = "second") %>%  
  autoplot() + labs(title = "Success rate curve: species richness")
```



```
roc_curve(simpson_success,  
          truth = simpson_class,  
          .pred_high,  
          event_level = "second") %>%  
  autoplot() + labs(title = "Success rate curve: Simpson diversity")
```



Summary Table : Success Rate vs Prediction Rate AUC

```
tibble::tibble(
  Outcome = c("Species richness", "Simpson diversity"),
  `Success rate AUC` = c(success_rate_richness$.estimate,
    success_rate_simpson$.estimate),
  `Prediction rate AUC` = c(overall_richness,
    overall_simpson),
  `Prediction rate SE` = c(overall_richness_metrics$std_err,
    overall_simpson_metrics$std_err)
) %>%
mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
knitr::kable(caption = "Random forest classification performance:
  success rate vs. prediction rate AUC")
```

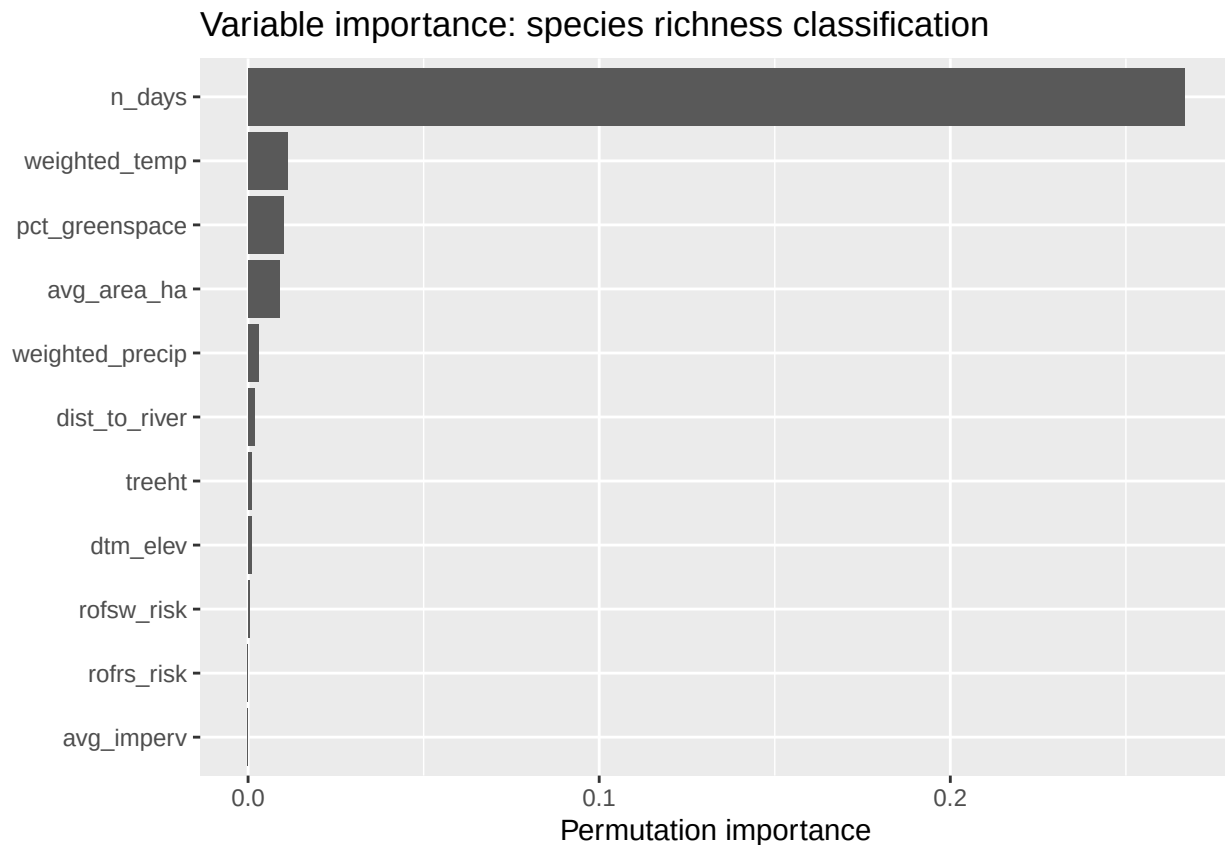
Table 4: Random forest classification performance: success rate vs. prediction rate AUC

Outcome	Success rate AUC	Prediction rate AUC	Prediction rate SE
Species richness	0.984	0.951	0.010
Simpson diversity	0.933	0.710	0.022

For the richness model, there is a relatively small decrease from the sample AUC (0.984) to the cross-city AUC (0.951), indicating that predictive performance remains similar when the model is applied to unseen cities. In the Simpson model, there is a larger decrease, 0.933 to 0.710, suggesting less consistent performance across cities.

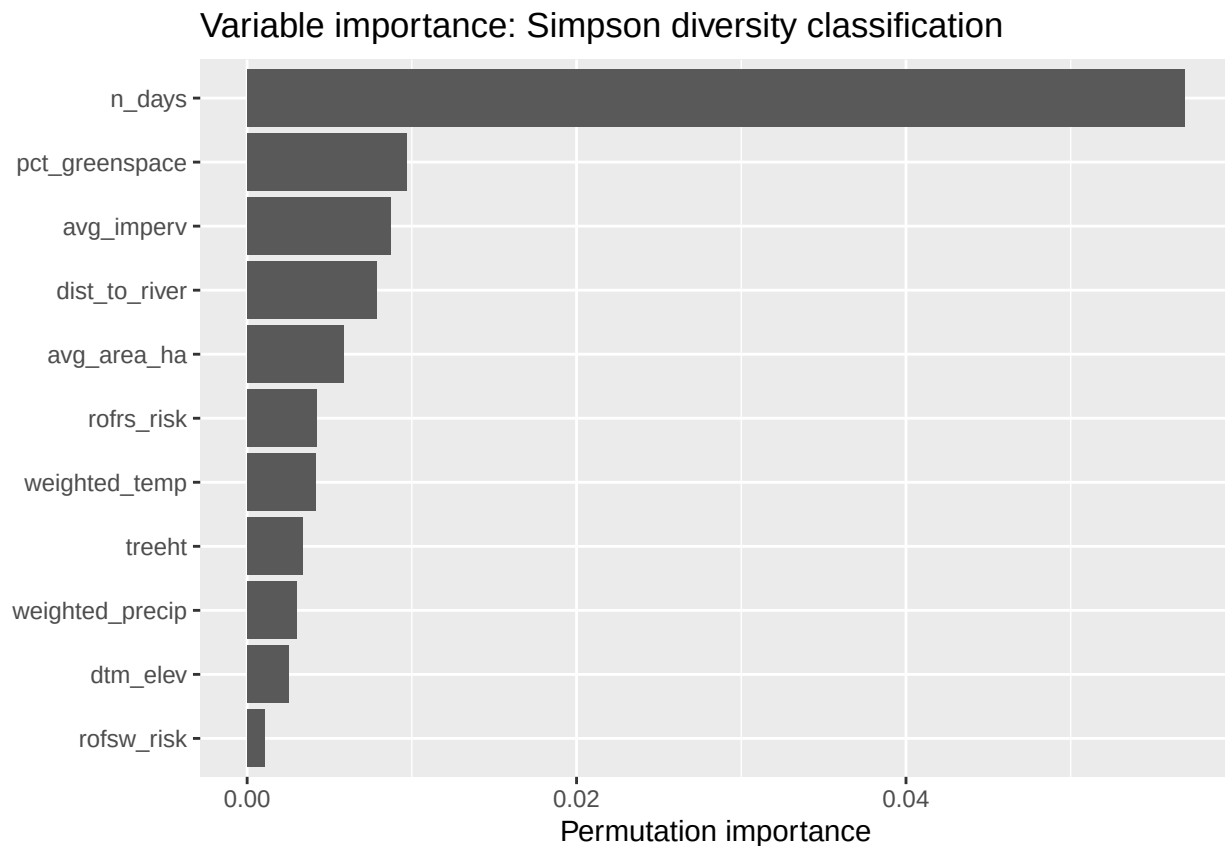
Variable Importance

```
richness_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
  as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col() +
  coord_flip() +
  labs(title = "Variable importance: species richness classification",
       x = NULL, y = "Permutation importance")
```



```
simpson_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
```

```
as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col() +
  coord_flip() +
  labs(title = "Variable importance: Simpson diversity classification",
       x = NULL, y = "Permutation importance")
```



Once again, **n_days** is the dominant predictor variable for both models. In the richness model, **weighted_temp**, **pct_greenpace**, and **avg_area_ha** show secondary importance. **weighted_temp** and **pct_greenpace** were also relevant in the regression random forest and significant in the negative binomial regression. In the Simpson model, **pct_greenpace**, **avg_imperv**, **dist_to_river**, and **avg_area_ha** show secondary importance.

The following notebook reruns these models without the predictor **n_days** to examine the relative importance of ecological drivers look after removing the dominant predictor sampling effort. It is a deliberate modelling decision expected to reduce model predictive performance. Removing **n_days** allows the relative importance of environmental predictors to become more apparent but doesn't necessarily provide a more accurate view of their ecological effects.